Apache Spark + MongoDB + PostgreSQL

Collaboration in a Cloud Environment:

… an integration scenario in the **Big Data** ecosystem, consisting in connecting a local computing environment (Spark in Colab) with a cloud database (MongoDB Atlas on AWS).

Apache Spark + MongoDB + PostgreSQL

Collaboration in a Cloud Environment:

The aim of the laboratory → is to connect various cloud environments: AWS, Colab with a local database system hosted in a Docker container.

we work with various data sets: Integration of data storage in Mongodb Atlas (AWS - hosted in Hamburg) with data processing and analysis in Apache Spark (Colab) and storing the results in a relational PostgreSQL database (Docker container)

Tasks:

1. Learn what MongoDB is and how to set up MongoDB cloud resources on AWS.

https://www.mongodb.com/

Provide access to the mongoDB database engine. To do this, configure access to Mongo Atlas on Cloud

https://www.mongodb.com/cloud/atlas/register

Create a new database and insert sample data into the database

Configure the environment for the Cloud:

Dataset for download: sample_mflix (movies, comments, users) or sample_analytics

https://www.mongodb.com/docs/atlas/sample-data/sample-mflix/

https://github.com/wakaidev/MFlix

2. In the MongoDB Atlas → cloud environment, define a new database for Apache Spark preprocessing

Understand document structure, calculate distributions, and write some aggregations in MongoDB. Results:

- Open MongoDB Atlas UI, browse collections and sample documents: sample_mflix
- Execute basic aggregate queries (number of movies per year, top genres, average rating if available).
- Write down three research questions (e.g., "which movies have the most comments?", "what is the distribution of transactions per customer?").

3. Downloading Data to Spark and Preprocessing

Connect Spark to MongoDB Atlas, load collections as DataFrames, and perform cleansing/transformation (e.g., denormalization, date field extraction, flattening).

→ Run a **SparkSession** in Colab with a parameter that adds the MongoDB Spark Connector (example).
→ Load the **movies** and **comments** collection into a DataFrame.
→ Join (e.g., append the number of comments to the movies), filter **movies** with more than X comments, and convert dates to timestamps.
→ Prepare aggregations (top 10 genres, rating distribution/long tail).

Apache Spark + MongoDB + PostgreSQL

Collaboration in a Cloud Environment:

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Tasks:

4. Transformations and Writing to a Relational Database

Process and write the resulting tables to a relational database (Postgres). Design simple table relationships (films, film_comments_agg, genres_stats).

→ Install the **Docker** environment
→ Search **DockerHub** and download the Docker container with the PostgreSQL database
→ Create a JDBC connection to Postgres (parameters: url, user, password).
→ Export the DataFrame(s) to SQL tables using DataFrame.write.jdbc(...).
→ Validate the data by performing a SELECT from Postgres (you can use sqlalchemy in Colab).

Table Structure (example):

FILMS (film_id PK, title, year, runtime, director)

FILM_COMMENTS_AGG (film_id FK, comments_count, avg_comment_length)

GENRES_STATS (genre, film_count, avg_comments_per_film)



MongoDB

VIDE – MongoDB documentation

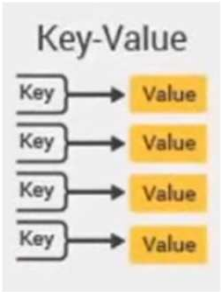
https://www.mongodb.com/

Questions:

What is the structure of relational databases and how is data stored in these structures?

How else can you store data in databases other than relational ones?

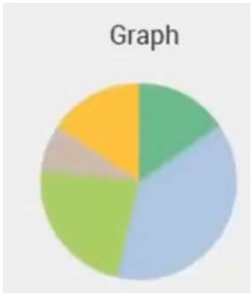
Databases



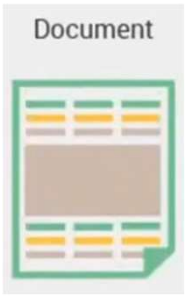
BerkeleyDB, LevelDB, Memcached, Project Voldemort, Redis, Riak.



The database used by one of the most popular social networking sites, Facebook, is based on this model. It is called Cassandra. Other examples of columnar databases include Hypertable, Hadoop, and Hbase.



FlockDB, HyperGraphDB, Infinite Graph, Neo4J, OrientDB



MongoDB and CouchDB, Orient DB and the IBM Domino database

https://www.mongodb.com/

https://db-engines.com/en/ranking

Leaders → are Multimodel DBMS!!!

MongoDB

BigData → Important!!!:

If you get questions (on the competency exam):

Why are **MultiModel** databases top of the list of all database engines?

What are **Multimodel Databases**, what are the benefits of using them, and how do they fit into Big Data processing? $→ you will be able to answer this!!!

424 systems in ranking, April 2025

Rank			DBMS	Database Model	Score		
Apr 2025	Mar 2025	Apr 2024			Apr 2025	Mar 2025	Apr 2024
1.	1.	1.	Oracle	Relational, Multi-model	1231.05	-22.03	-3.21
2.	2.	2.	MySQL	Relational, Multi-model	987.11	-1.02	-100.62
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model	785.01	-3.13	-44.79
4.	4.	4.	PostgreSQL	Relational, Multi-model	667.25	+3.82	+22.20
5.	5.	5.	MongoDB	Document, Multi-model	400.05	+3.63	-23.91
6.	6.	↑ 9.	Snowflake	Relational	168.09	+6.31	+44.89
7.	7.	↓ 6.	Redis	Key-value, Multi-model	154.11	-1.25	-2.33
8.	8.	↓ 7.	Elasticsearch	Multi-model	128.08	-3.30	-6.70
9.	9.	↓ 8.	IBM Db2	Relational, Multi-model	126.04	-0.53	-1.45
10.	10.	10.	SQLite	Relational	114.09	+1.01	-1.92
11.	11.	↑ 12.	Apache Cassandra	Wide column, Multi-model	108.21	+1.55	+4.34
12.	↑ 13.	↑ 17.	Databricks	Multi-model	99.29	+3.28	+22.96
13.	↓ 12.	↓ 11.	Microsoft Access	Relational	95.36	-1.35	-10.04
14.	14.	↓ 13.	MariaDB	Relational, Multi-model	94.34	+0.11	+0.53
15.	↑ 16.	↑ 16.	Amazon DynamoDB	Multi-model	76.49	-0.08	-1.08
16.	↑ 17.	↓ 15.	Microsoft Azure SQL Database	Relational, Multi-model	75.37	+1.46	-3.03
17.	↓ 15.	↓ 14.	Splunk	Search engine	73.38	-5.49	-15.33
18.	18.	18.	Apache Hive	Relational	72.65	+2.57	+10.07
19.	19.	19.	Google BigQuery	Relational	60.73	+2.77	-1.17
20.	20.	↑ 23.	Neo4j	Graph	47.59	+1.44	+3.12

1. MongoDB (from English: humongous[1]):

- open,
- non-relational

database management system written in C++.

It is characterized by the lack of a precisely defined structure of the supported databases.

… data → is stored as JSON-style documents.

Possibilities:

- support for the Unicode standard (UTF),
- handling data in other encodings in binary format,
- a large number of supported data types,
- cursor support,
- ad-hoc queries,
- queries to nested document fields,

- indexing,
- support for data aggregation,
- the ability to store files in the database,
- architecture designed for easy replication.

The internal language for defining queries and aggregate functions is JavaScript

Laboratory:

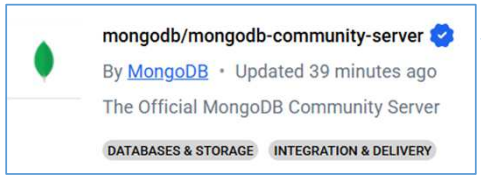
→ Tasks: Spring + Apache Spark

A. Provide access to the database engine: mongoDB

for this purpose:

in resources:

docker hub



or

Configure access to Mongo Atlas on Cloud

Setup a Cluster in MongoDB Atlas

https://www.youtube.com/watch?v=0SAgAIBYXTY

A. MongoDB Atlas cluster configuration: in the Google browser we enter: cloud+mongodb+atlas

for this purpose: https://www.mongodb.com/cloud/atlas/register

…then → …after answering the survey we choose → Free



Deploy your cluster

Use a template below or set up advanced configuration options. You can also edit these configuration options once the cluster is created.

M10 $0.09/hour

Dedicated cluster for development environments and low-traffic applications.

STORAGE	RAM	vCPU
10 GB	2 GB	2 vCPUs

Flex From $0.011/hour
Up to $30/month

For application development and testing, with on-demand burst capacity for unpredictable traffic.

STORAGE	RAM	vCPU
5 GB	Shared	Shared

Free

For learning and exploring MongoDB in a cloud environment.

STORAGE	RAM	vCPU
512 MB	Shared	Shared

Free forever! Your free cluster is ideal for experimenting in a limited sandbox. You can upgrade to a production cluster anytime.

Configurations

Name
You cannot change the name once the cluster is created.

Quick setup

- [x] Automate security setup
- [x] Preload sample dataset

I'll do this later | Go to Advanced Configuration | Create Deployment

docker hub

A. MongoDB Atlas cluster configuration:

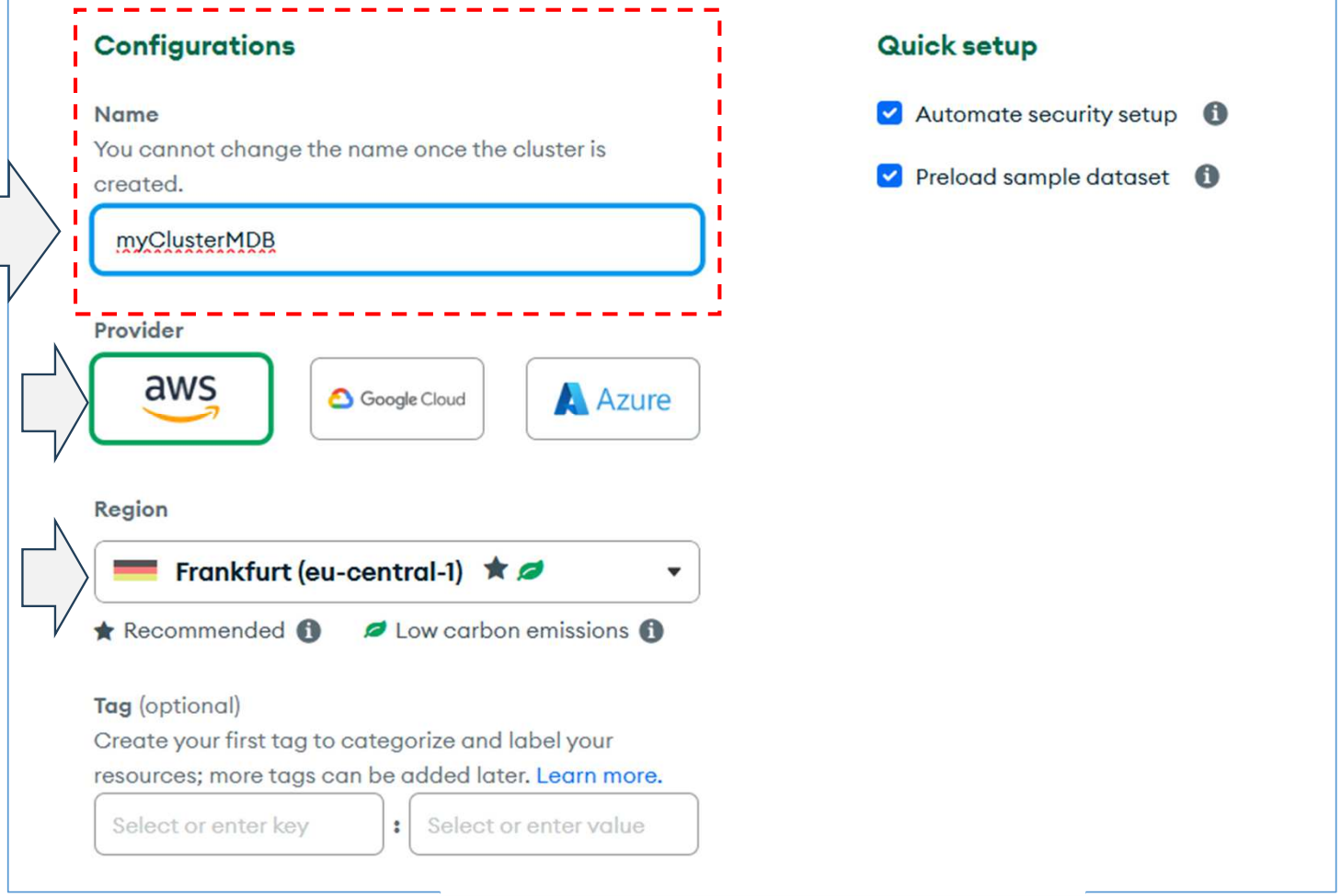
…then →

… insert the name of our cluster :

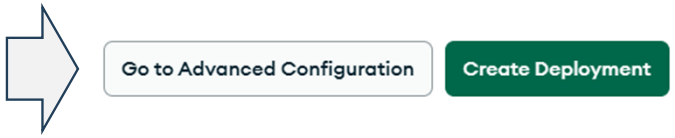
myClusterMDB

…set the Cloud provider for the cluster, e.g. AWS:

…we set the cluster location



… choose to create a cluster



A. MongoDB Atlas cluster configuration:

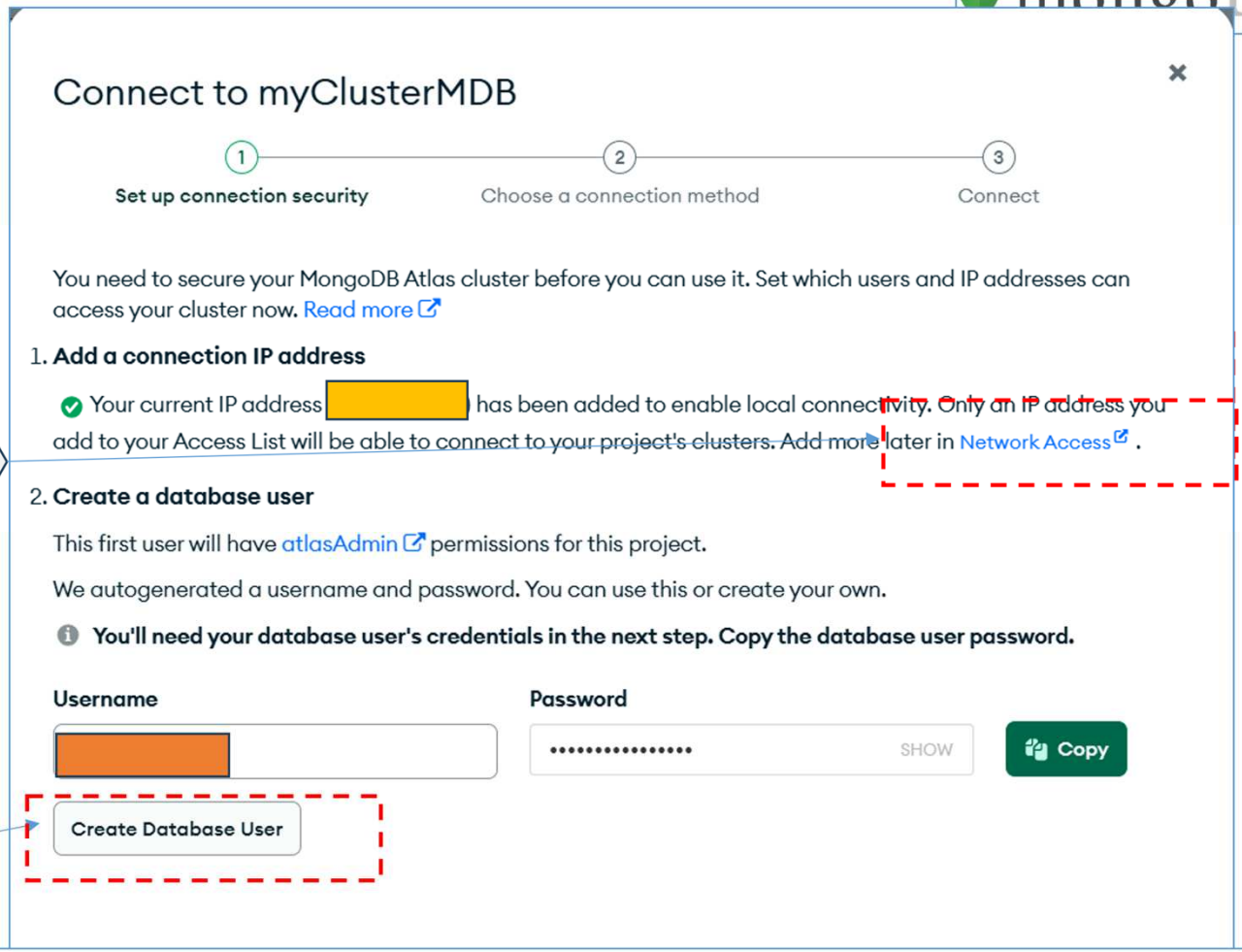
… then →

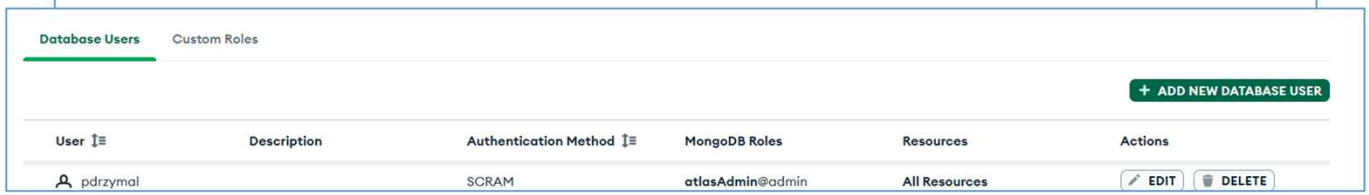
… write down the IP name from which the connection can be made:

… and add any IP from which the connection can be made:

… set up an additional user :
BIGDATA
BigData

… define a new cluster user





A. MongoDB Atlas cluster configuration:

...next →



Add New Database User

Create a database user to grant an application or user access to databases and collections in your clusters in this Atlas project. Granular access control can be configured with default privileges or custom roles. You can grant access to an Atlas project or organization using the corresponding Access Manager

Authentication Method

... set up an additional user :
BIGDATA
BigData

Password | Certificate | AWS IAM | Federated Auth (MongoDB 7.0 and up)

MongoDB uses SCRAM as its default authentication method.

Password Authentication

BIGDATA

BigData HIDE

Autogenerate Secure Password | Copy

User Description

Add an optional description to your user.

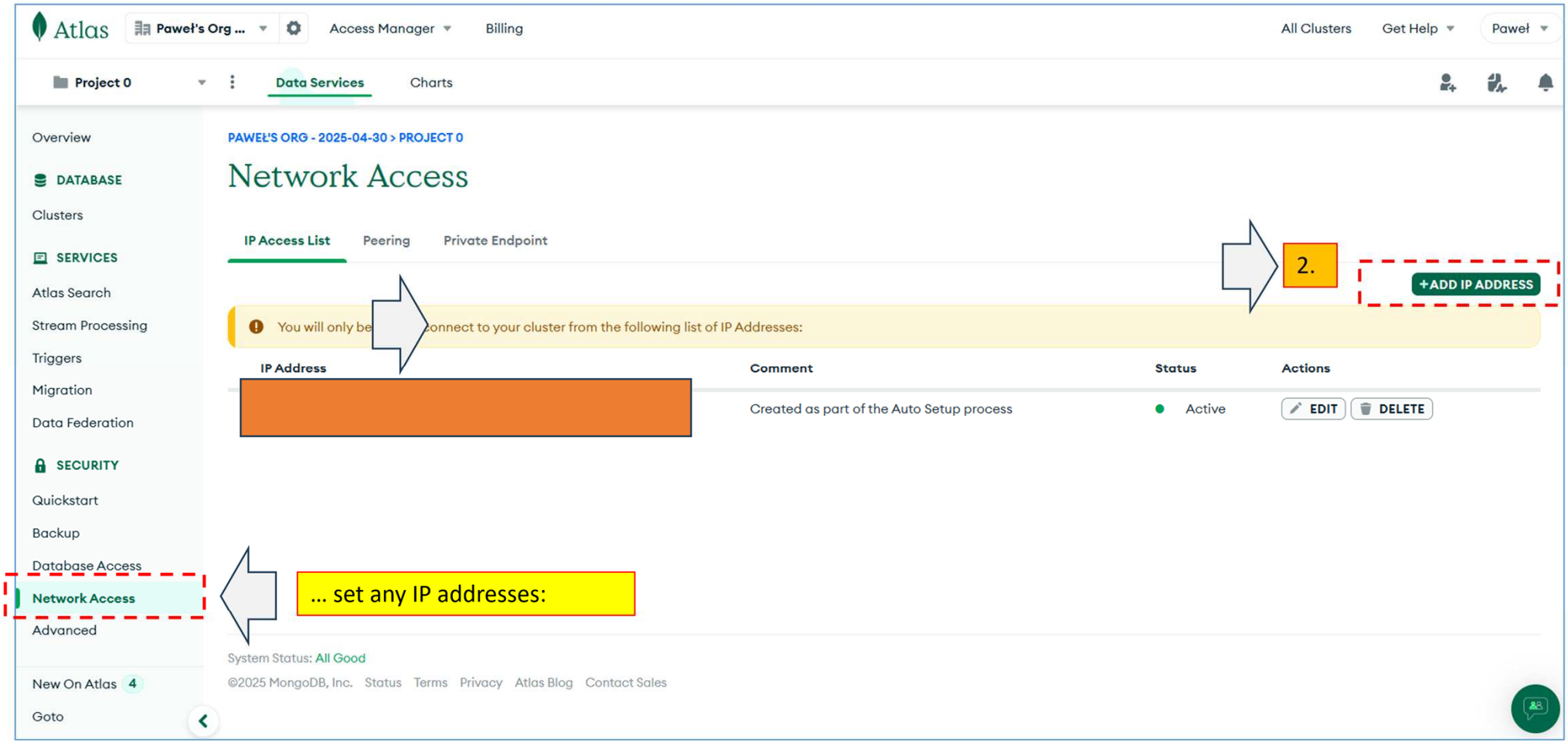
... We choose to create a cluster

Go to Advanced Configuration | Create Deployment

A. MongoDB Atlas cluster configuration:

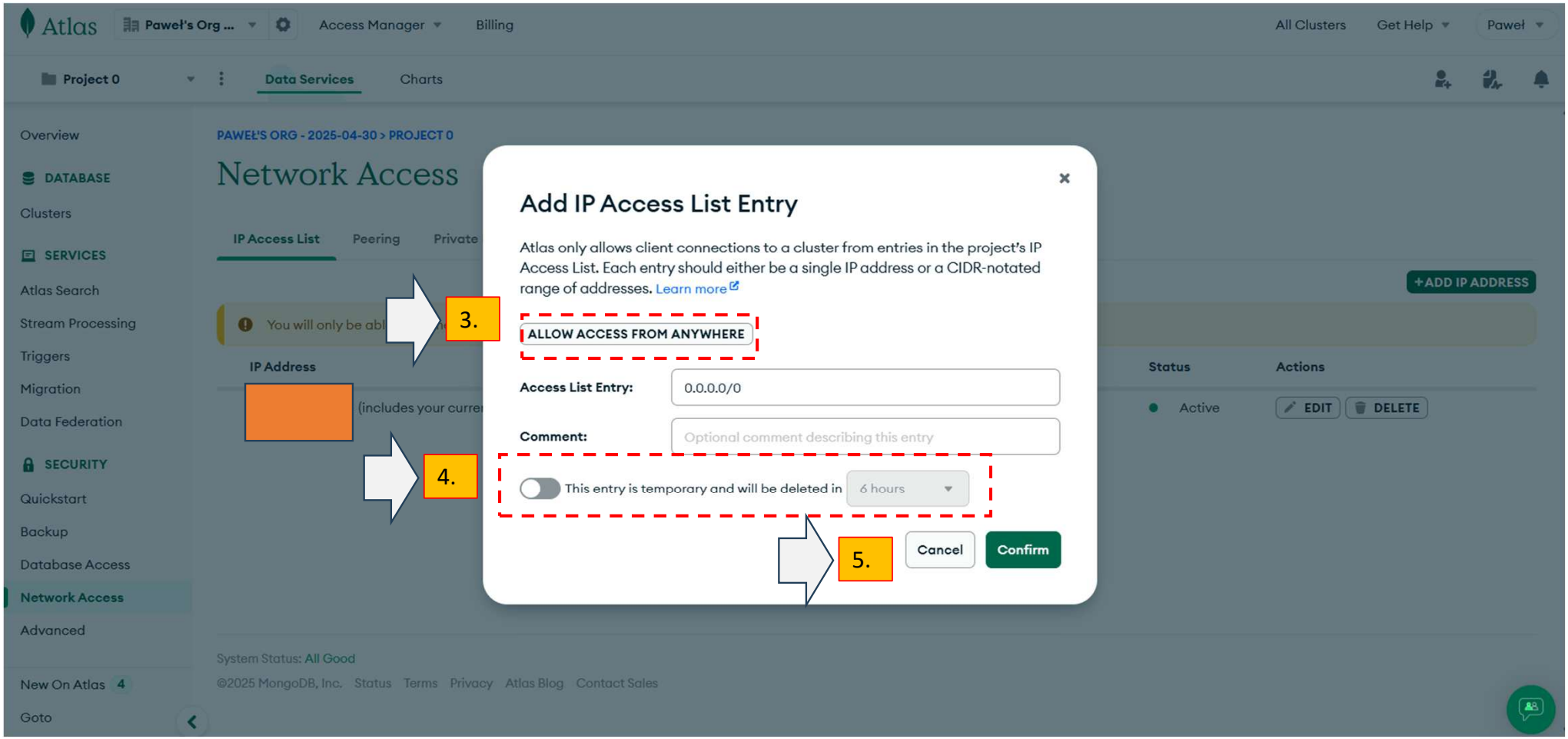
...next →

... set any IP addresses:



A. MongoDB Atlas cluster configuration:

…next →



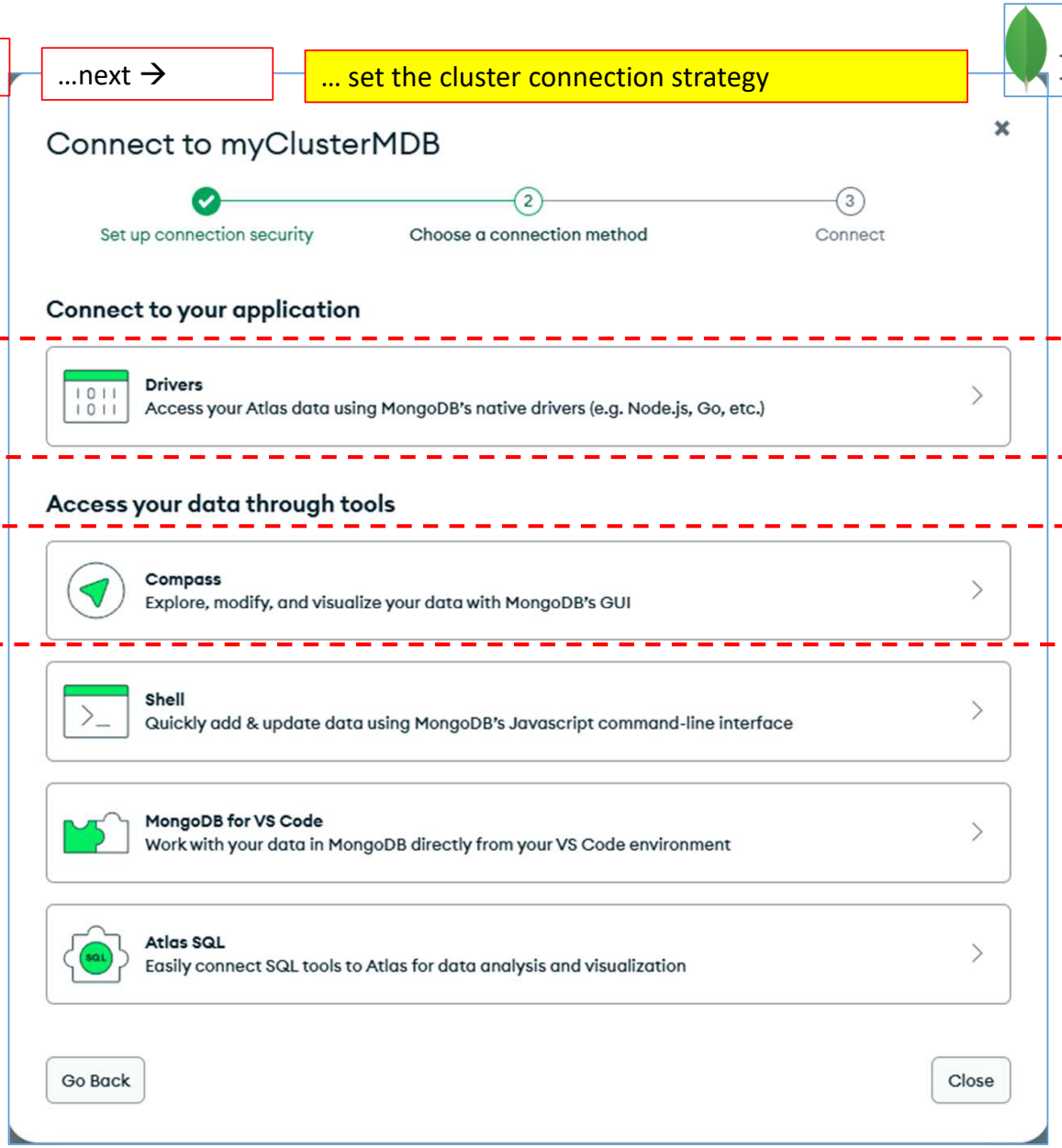
A. MongoDB Atlas cluster configuration:

…next →

… set the cluster connection strategy

… select the connection to the cluster → 6.

… +optional (in the second step we select the connection to the cluster → 7.



A. MongoDB Atlas cluster configuration:

…next →

… set the cluster connection strategy

… select the connection to the cluster

8.

ATTENTION !!!

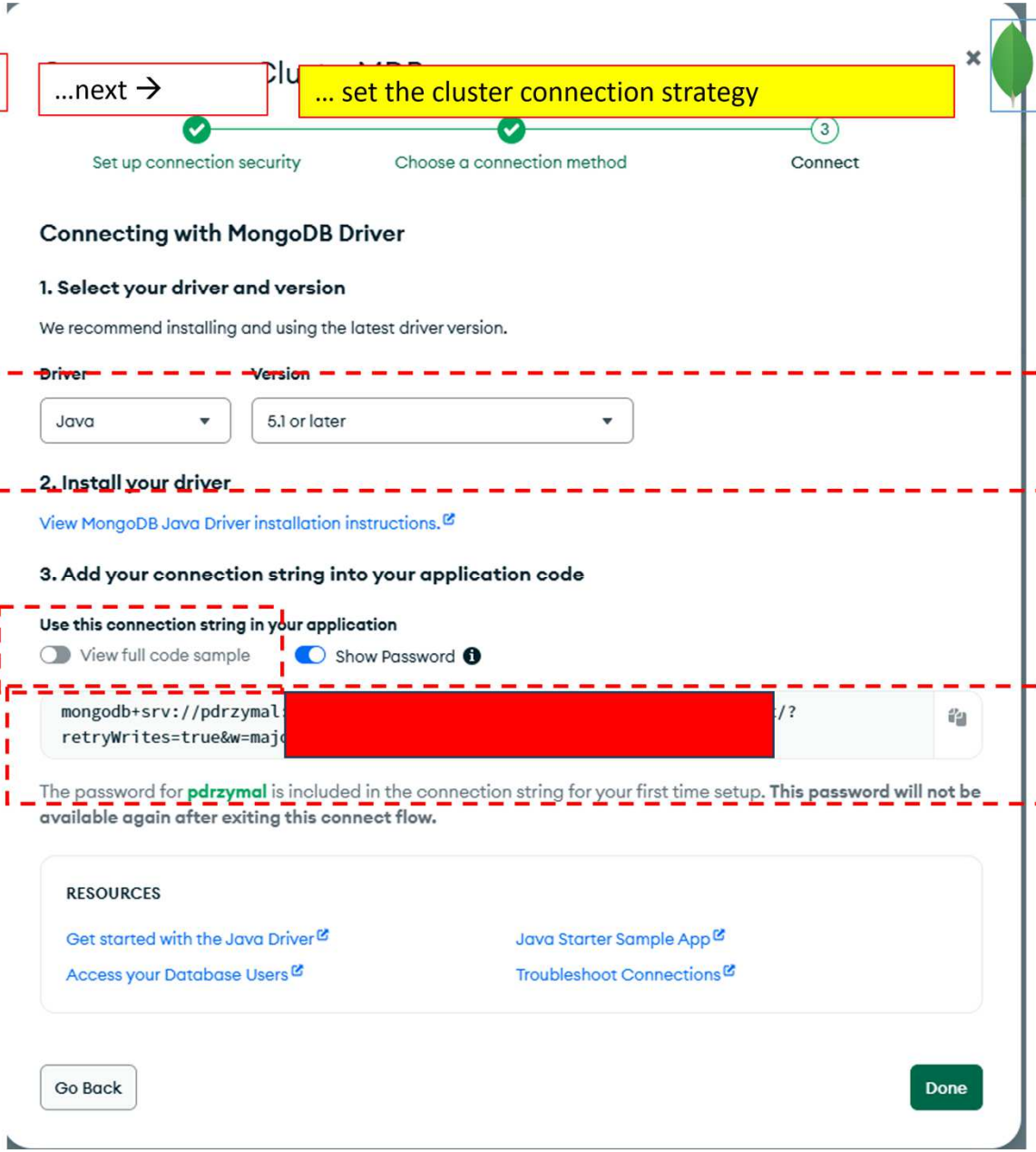
… select and copy

ATTENTION !!!

… and copy the connection string

9.

Even though the password is one-time use, we have a url



A. MongoDB Atlas cluster configuration:

… set the cluster connection strategy

2. Install your driver

View MongoDB Java Driver installation instructions.

3. Add your connection string into your application code

Use this connection string in your application

View full code sample Show Password

```
import com.mongodb.ConnectionString;
import com.mongodb.MongoClientSettings;
import com.mongodb.MongoException;
import com.mongodb.ServerApi;
import com.mongodb.ServerApiVersion;
import com.mongodb.client.MongoClient;
import com.mongodb.client.MongoClients;
import com.mongodb.client.MongoDatabase;
import org.bson.Document;

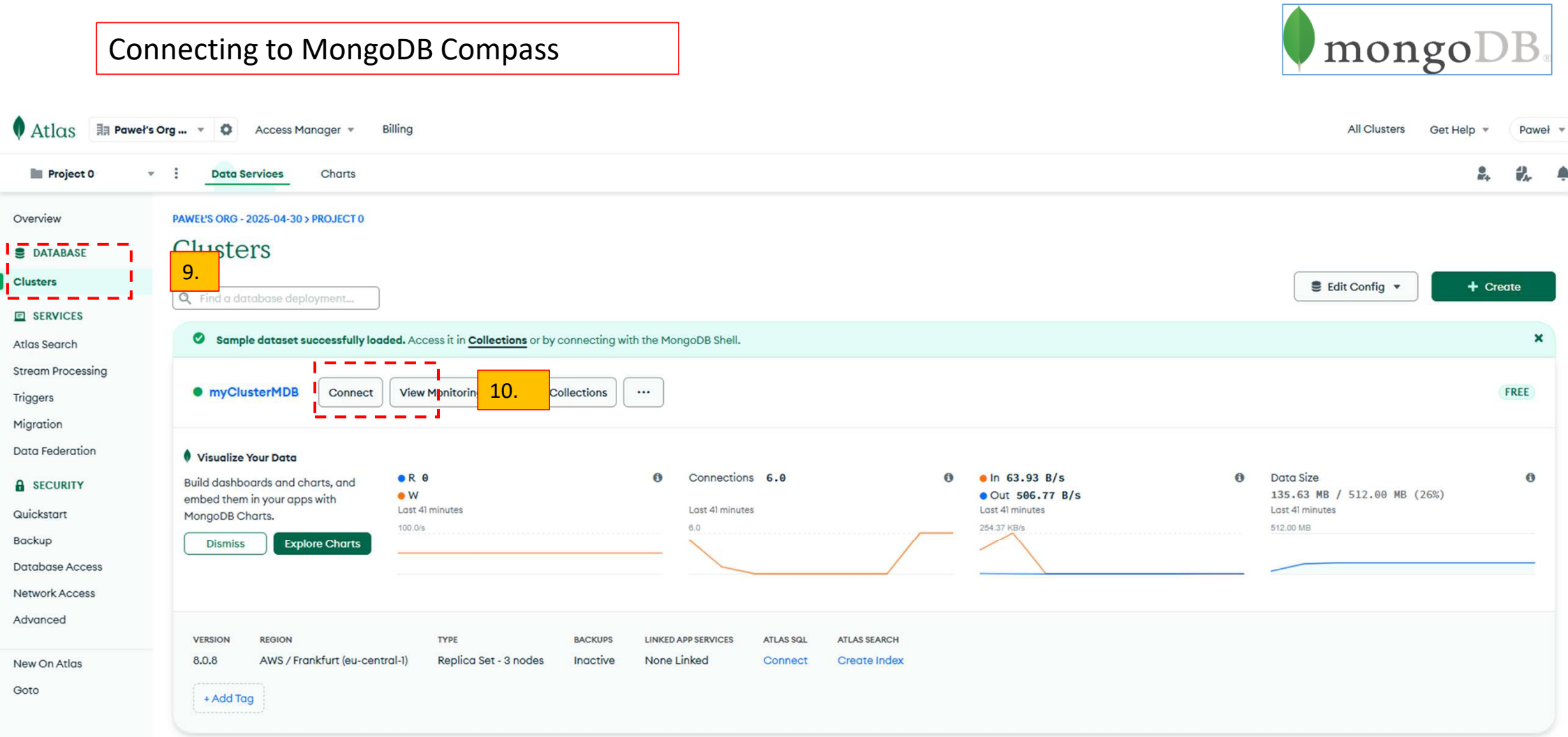
public class MongoClientConnectionEx
    public static void main(String[]
        String connectionString = "m                                                    db.

        ServerApi serverApi = ServerApi.builder()
                .version(ServerApiVersion.V1)
                .build();

        MongoClientSettings settings = MongoClientSettings.builder()
                .applyConnectionString(new ConnectionString(connectionString))
                .serverApi(serverApi)
                .build();

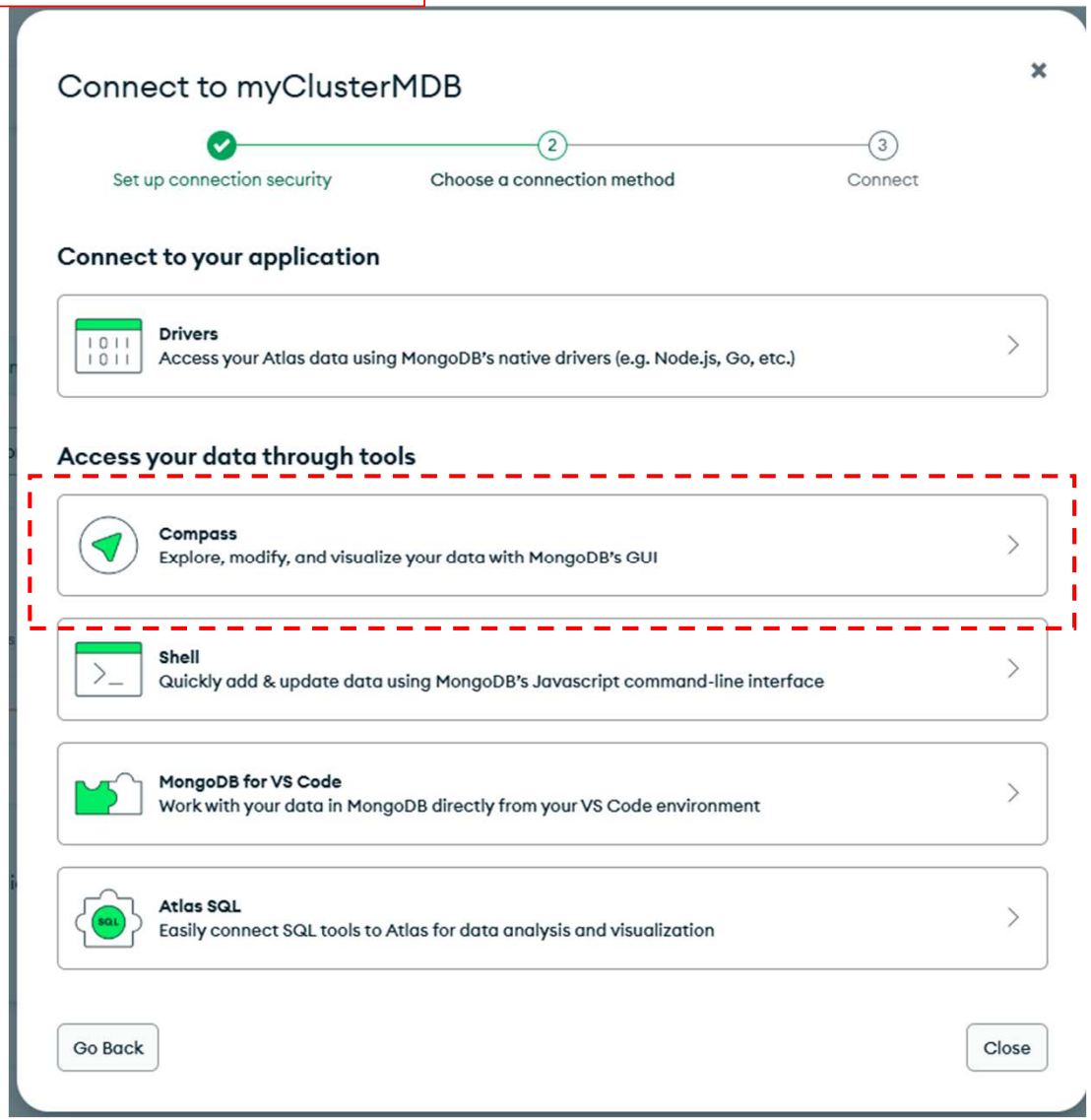
        // Create a new client and connect to the server
        try (MongoClient mongoClient = MongoClients.create(settings)) {
            try {
                // Send a ping to confirm a successful connection
                MongoDatabase database = mongoClient.getDatabase("admin");
                database.runCommand(new Document("ping", 1));
                System.out.println("Pinged your deployment. You successfully connected t
            } catch (MongoException e) {
                e.printStackTrace();
            }
        }
    }
}
```

Connecting to MongoDB Compass

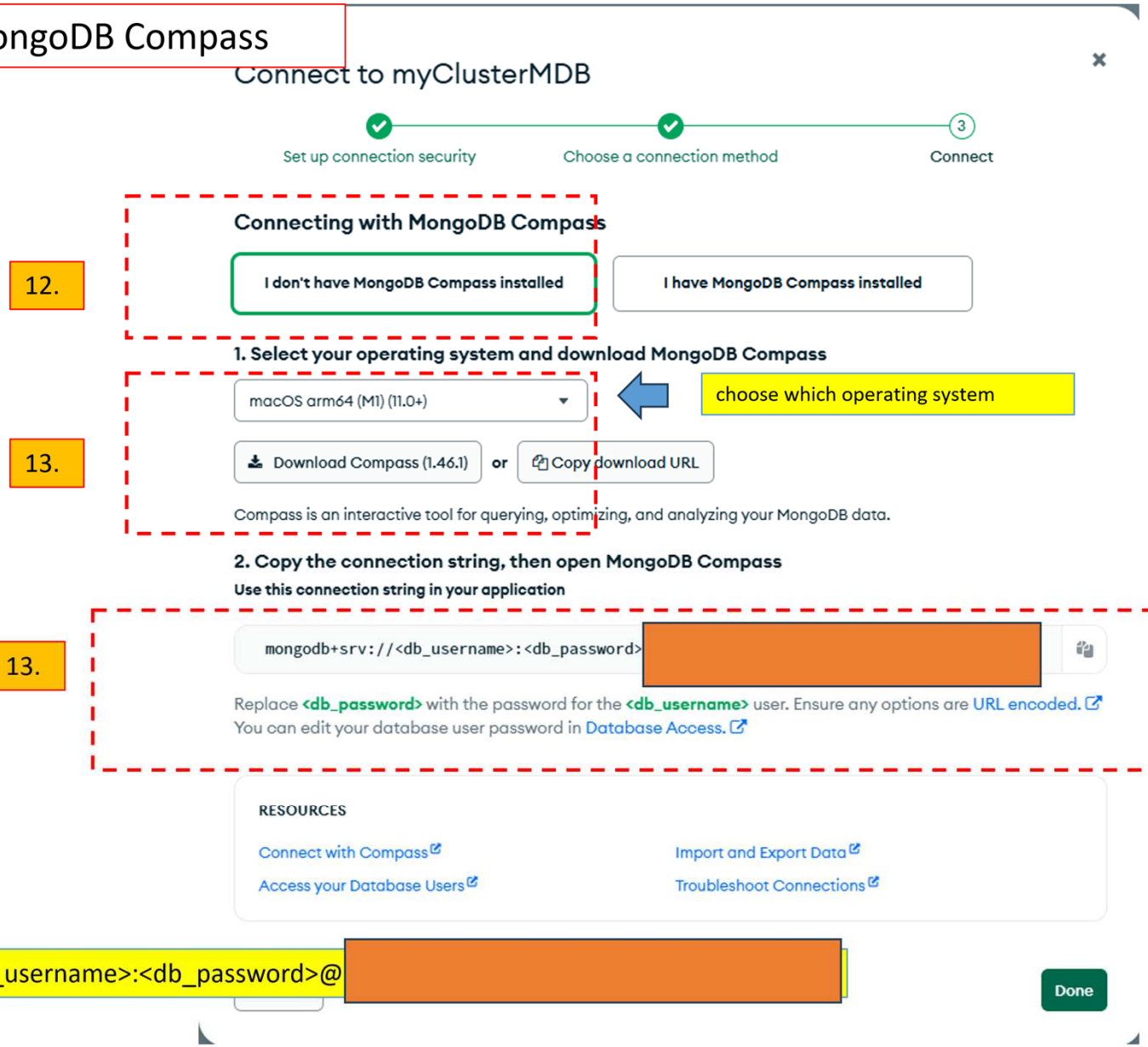


Connecting to MongoDB Compass

11.



Connecting to MongoDB Compass



mongodb+srv://<db_username>:<db_password>@

MongoDB Compass

Connections Edit View Help

Connecting to MongoDB Compass

Compass

My Queries

CONNECTIONS (1)

Search connections

[redacted]

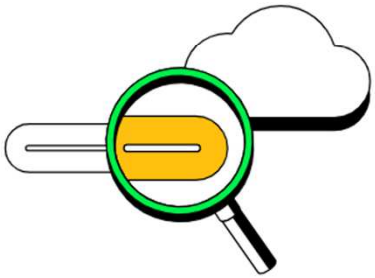
admin

local

sample_mflix

13.

Welcome



Welcome to MongoDB Compass

To get started, connect to an existing server or

+ Add new connection

New to Compass and don't have a cluster?

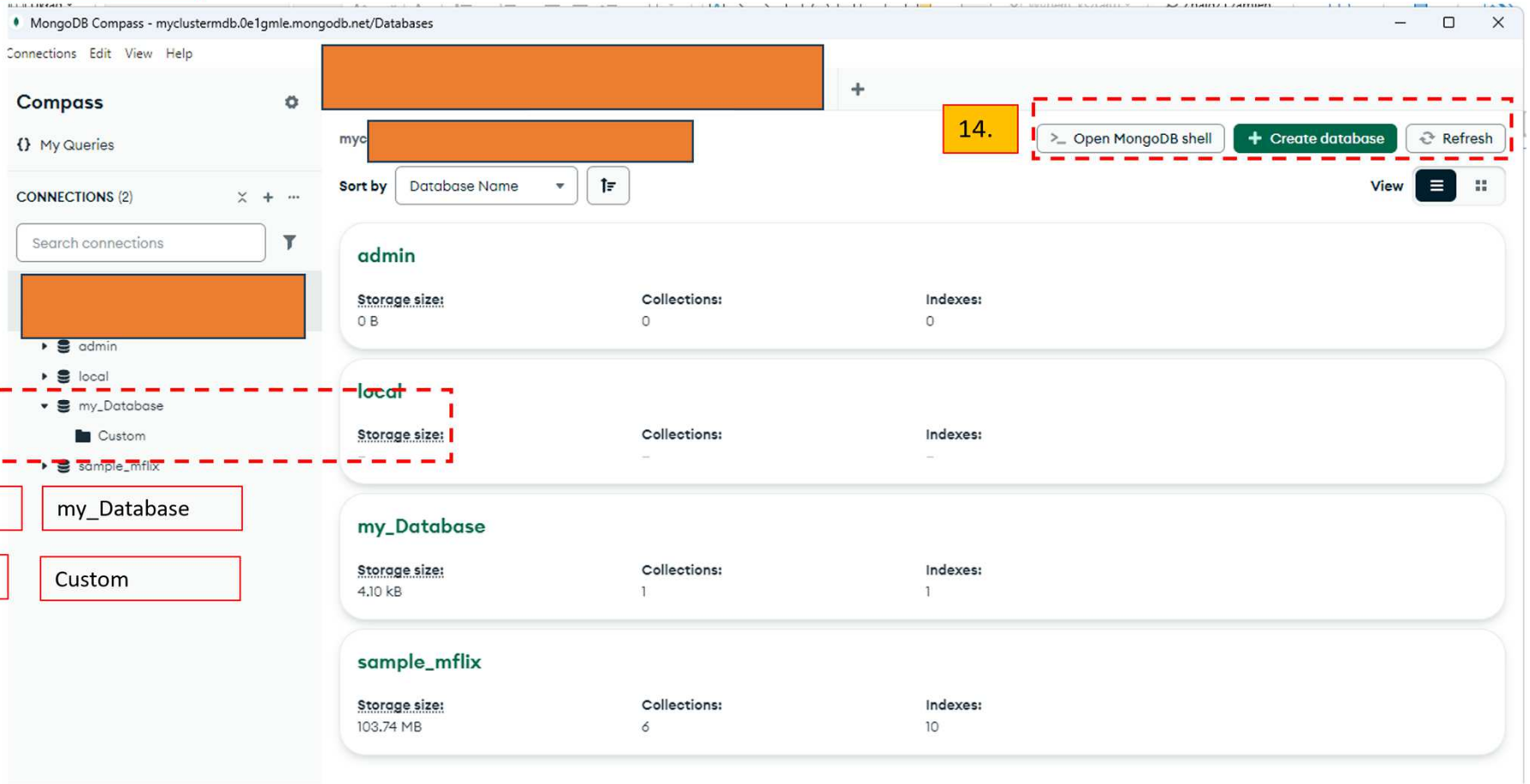
If you don't already have a cluster, you can create one for free using MongoDB Atlas

CREATE FREE CLUSTER

mongodb+srv://<db_username>:<db_password>@[redacted]

Connecting to MongoDB Compass

… add a database:



15.

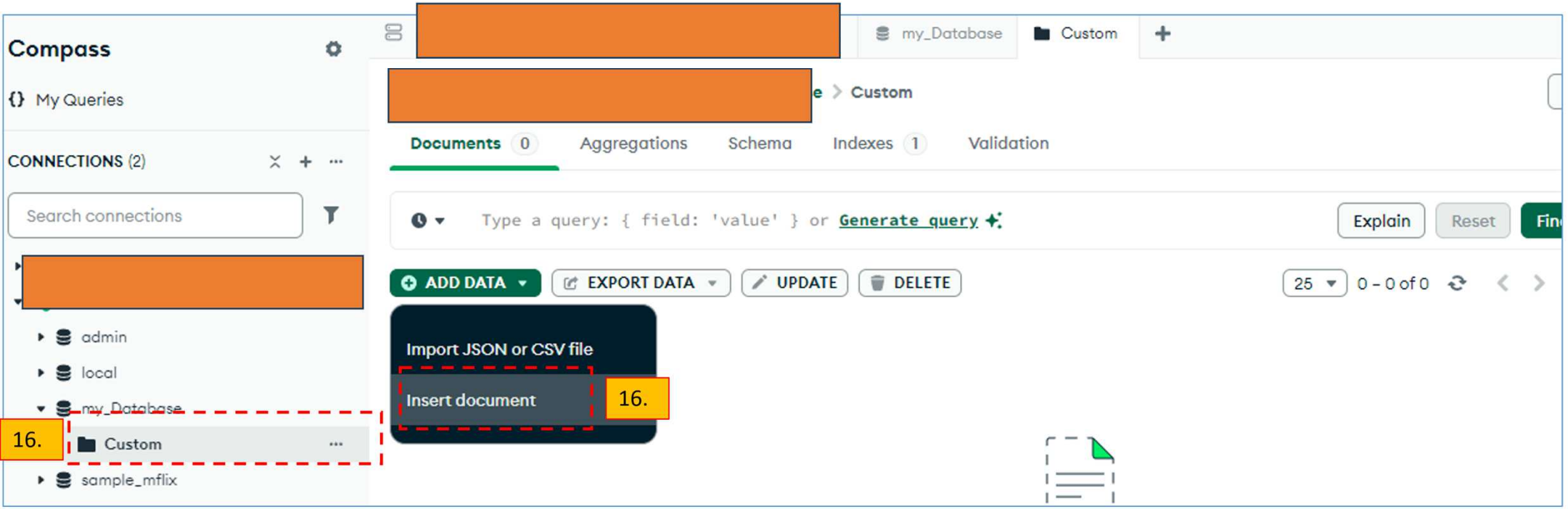
Database: my_Database

Schema: Custom

mongodb+srv://<db_username>:<db_password>@

Connecting to MongoDB Compass

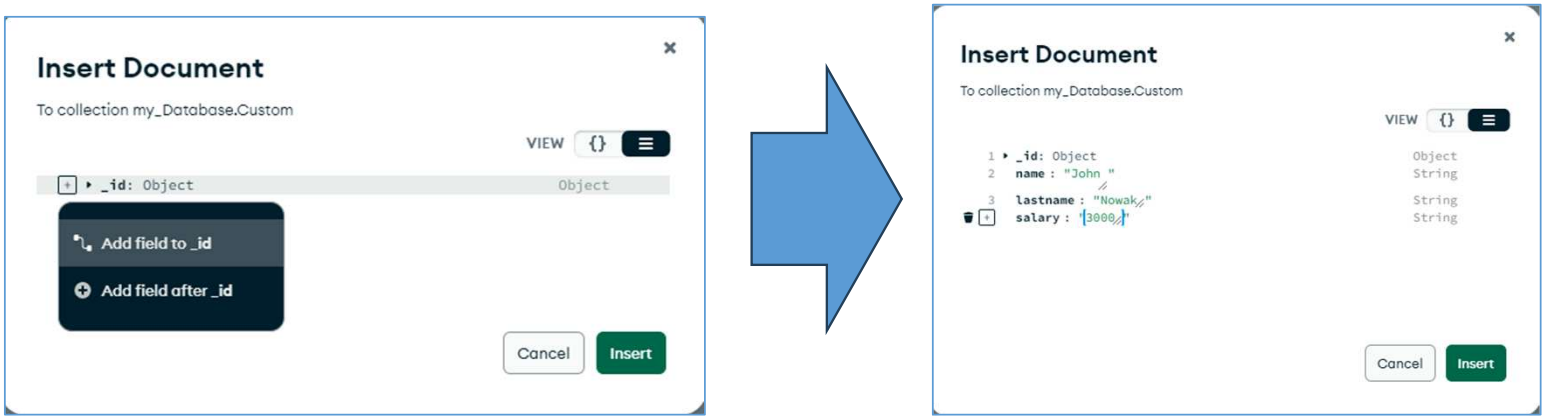
data insertion:

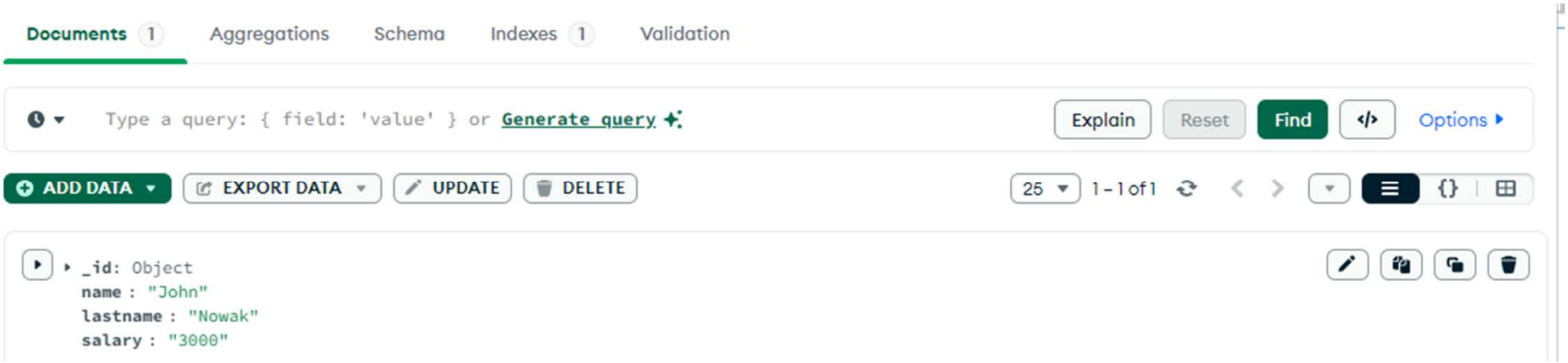


mongodb+srv://<db_username>:<db_password>@

Connecting to MongoDB Compass

data insertion:





Connecting to MongoDB Compass

data insertion:

